Risk Fingerprints

Where the INFORM Risk Index misses

A custom interactive visualization of prediction versus reality

Two views of disaster risk

A model predicts. Reality records. **The gap is the finding.**

THE PREDICTION

INFORM Risk Index 2026

A composite score from 0 to 10 for every country. Combines hazard exposure, population vulnerability, and lack of coping capacity. Joint Research Centre, European Commission. CC BY 4.0.

THE REALITY

EM-DAT, 1995 to 2025

Event level records of natural disasters with deaths, people affected, and damages. 10,759 events across 191 countries. Centre for Research on the Epidemiology of Disasters, UCLouvain.

Assignment 2, and what the grader told us

Power BI dashboard, four pages, scored 26 out of 30. The feedback was specific and we designed A3 to answer it.

WHAT THE GRADER SAID		WHAT A3 DOES
Filter behavior across pages was clunky.	→	**Instant, client side linked views.**
Scatter colored by country, 191 unreadable colors.	→	**Scatter colored by region, five colors.**
Plans for next steps were unclear.	→	**This whole project is the next step.**

Brief also requires going beyond standard chart libraries. **Everything here is hand built with D3 and SVG. No Plotly, no Recharts, no Chart.js.**

The residual is the visualization

For each country we compute the gap between what the model predicted and what actually happened.

$$\textbf{residual} = \log_{10}(\text{actual deaths} + 1) - \text{predicted by OLS on INFORM risk}$$

positive residual $\longrightarrow$ *more deaths than predicted* · *negative residual* $\longrightarrow$ *fewer deaths than predicted*

The whole main page is sorted by this single number, top to bottom.

A viewer should grasp “this grid is sorted by model error” within five seconds.

DESIGN PRINCIPLE

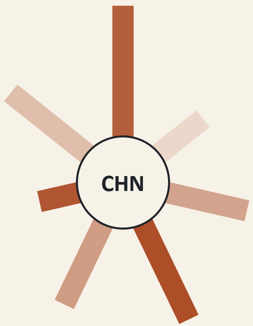
The sort order is not metadata. The sort order is the finding.

User tasks the visualization supports

	DOMAIN TASK	ABSTRACTED
T1	Find where INFORM over or under predicts a specific hazard.	*Identify outliers*
T2	Compare two or more countries' hazard fingerprints.	*Compare multivariate items*
T3	Examine the actual events behind a flagged country's losses.	*Drill down to records*

Per the brief, these are the user tasks the visualization supports, not the development tasks we performed.

The petal glyph: prediction and reality on one mark



a single country's fingerprint

ONE GLYPH PER COUNTRY

Seven hazards, fixed angular order

River flood, coastal flood, earthquake, tropical cyclone, drought, tsunami, epidemic. Same hazard sits in the same place on every country, so shapes are directly comparable.

Petal length = INFORM risk score (0–10) *the prediction*

Petal saturation = $\log_{10}$ deaths from that hazard *the reality*

WHY THIS DESIGN

Shape recognition across small multiples is faster than scanning 191 separate bar charts.

Dual encoding on one mark puts prediction and reality together; a long pale petal vs. a short deep petal reads instantly.

Four coordinated views

1

Fingerprint grid

All 191 countries as radial glyphs. Sorted top to bottom by signed residual, with an explicit “expected” divider line. The sort is the finding.

2

Scatter with residual sticks

INFORM risk x log deaths. Region colored, not country. Explicit OLS line plus a 1 and 2-sigma prediction band. Each dot has a stick drawn to the line, literally showing the quantity the grid is sorted by.

3

Event timeline

When a country is selected, that country's actual EM-DAT events drawn one circle per event over 1995 to 2025, colored by hazard. The drill that T3 asks for.

4

Hazard streamgraph

Deaths by hazard per year. Brush a year window and every residual is recomputed for just that era. The grid re-sorts. Focus plus context.

Interactions, all instant after one data load

Hover, click, brush linking

Selection in any view filters the others instantly.

Focus a hazard chip

Re-sorts grid and scatter by that single hazard's own OLS residual. Serves T1.

Region legend, isolating clicks

Click a region color to isolate it across both scatter and grid.

Pin to compare

Stack enlarged fingerprints side by side. Serves T2.

Country search, / to focus

By name or ISO3.

Time window brush

Drag on streamgraph; residuals recompute for that range and grid re-sorts.

Timeline legend filter

Click a hazard chip on the timeline to filter events.

Esc / Clear

Resets selection and filters.

Resizable split panel

Reader controls how much screen the grid versus the right pane gets.

Deep linkable URL state

Hazard, selection, year window, pins all encoded in the URL.

?years=2004-2005 & hazard=tsunami

What the visualization surfaces

THE EXTREMES

+2.73

China is the most under predicted country. INFORM scored it 2.9, yet about 115,000 deaths.

−2.97

Eritrea is the most over predicted. High INFORM score, few recorded deaths.

Beyond two sigma the misses are statistical, not noise. There are nine of them.

THE STRUCTURE

The error is not random.

Europe is systematically under predicted. Median residual +0.97, and 83% of European countries had more deaths than the model expected. The 2003 and 2010 heatwaves dominate.

Oceania is systematically over predicted. Median residual −0.86.

The error is era dependent. Brush to 2003 to 2007 and the top misses become Indonesia, France, Italy, Spain, Sri Lanka. One tsunami plus one heatwave.

A data quality fix that became a finding

EM-DAT classifies disasters as Type followed by Subtype. The Type level hides hazards we needed to keep separate.

BEFORE

Earthquake type, no subtype split

~ 253,000 tsunami deaths hidden inside the earthquake petal

AFTER

Subtype extracted, tsunami split out

Indonesia 171k, Sri Lanka 35k, Japan 20k now visible as the tsunami petal

This is exactly the kind of finding the project is designed to produce.

Looking at the grid we noticed Indonesia and Japan should have been more under predicted than they were. *Tracing back why led us into the data and uncovered ~253,000 deaths that the default Type field was hiding.* **The visualization made the question askable.**

What we deliberately scoped out

Natural hazards only

INFORM weights human and conflict hazards heavily, but EM-DAT carries no conflict outcome data. With no reality to measure prediction against, we scope conflict out of the main visualization on purpose. we do address it as a bonus, separately and clearly framed.

Coastal flood and tsunami required a subtype split

EM-DAT does not surface these at the Type level. The split is documented in the project and shown explicitly in the previous slide.

The scoping is the argument.

Choosing to measure where you can be measured, and being clear about where you cannot, is itself a design decision.

How it was built

FRONTEND	Next.js 14 (App Router), TypeScript, Tailwind for layout and type only
VISUALIZATION	D3 v7 plus hand built SVG. No chart libraries.
DATA PIPELINE	Node build step joins INFORM and EM-DAT into one static JSON, served from /public
DEPLOYMENT	Fully dockerized, multi stage build, Next standalone output
STATE	View state in the URL; every selection is a shareable link

LIVE

https://aiv.krenarahmeti.com

SOURCE

github.com/outreach-al/advanced-interactive-visualization

Reflection and what we would do next

CUSTOM VIS VS POWER BI

Power BI gave us the dataset, the join, the slicers. **It could not give us the residual sort, the dual encoded glyph, or the instant linking.** Going custom let us design around the finding rather than around the chart types a library happened to offer.

WHAT THE COST WAS

Far more time on layout, transitions, and accessibility than on the analysis. *Every alignment and color was a choice we had to make and defend.* Power BI absorbs those decisions for you. Doing them yourself is the cost of being able to design freely.

Where we would take it next

Generalize the encoding (any predicted vs observed pair, not just disasters), add a small-multiples comparison view for several hazards at once, and surface model confidence per country.

BONUS

Beyond the brief, a separate look at conflict

/conflict, unlinked from the main visualization

What it shows

An animated world choropleth across 2017 to 2026 with pan and zoom, plus a heatmap of INFORM's Violent Conflict Probability score. Framed as: did the model see it coming? Haiti and Ecuador rising sharply, Ukraine, the Sahel.

Why it is separate from the main view

There is no conflict outcome data in our dataset. So there is deliberately no residual. This page is prediction only, and that limitation is exactly why it lives on its own page and is not folded into the fingerprint grid.

VISUAL: screenshot of /conflict choropleth + Violent Conflict Probability heatmap

Thank you

Questions, or a live walkthrough